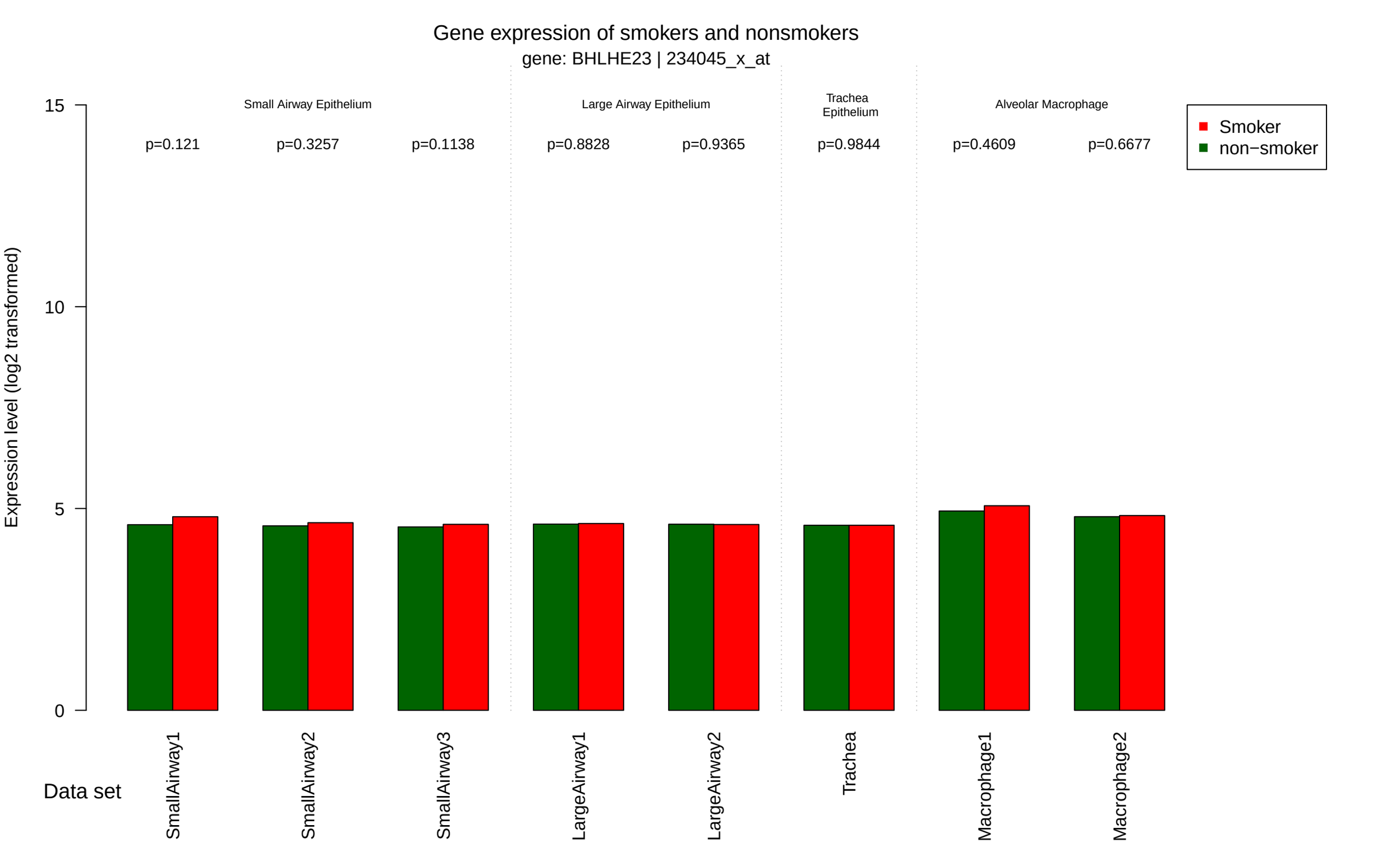
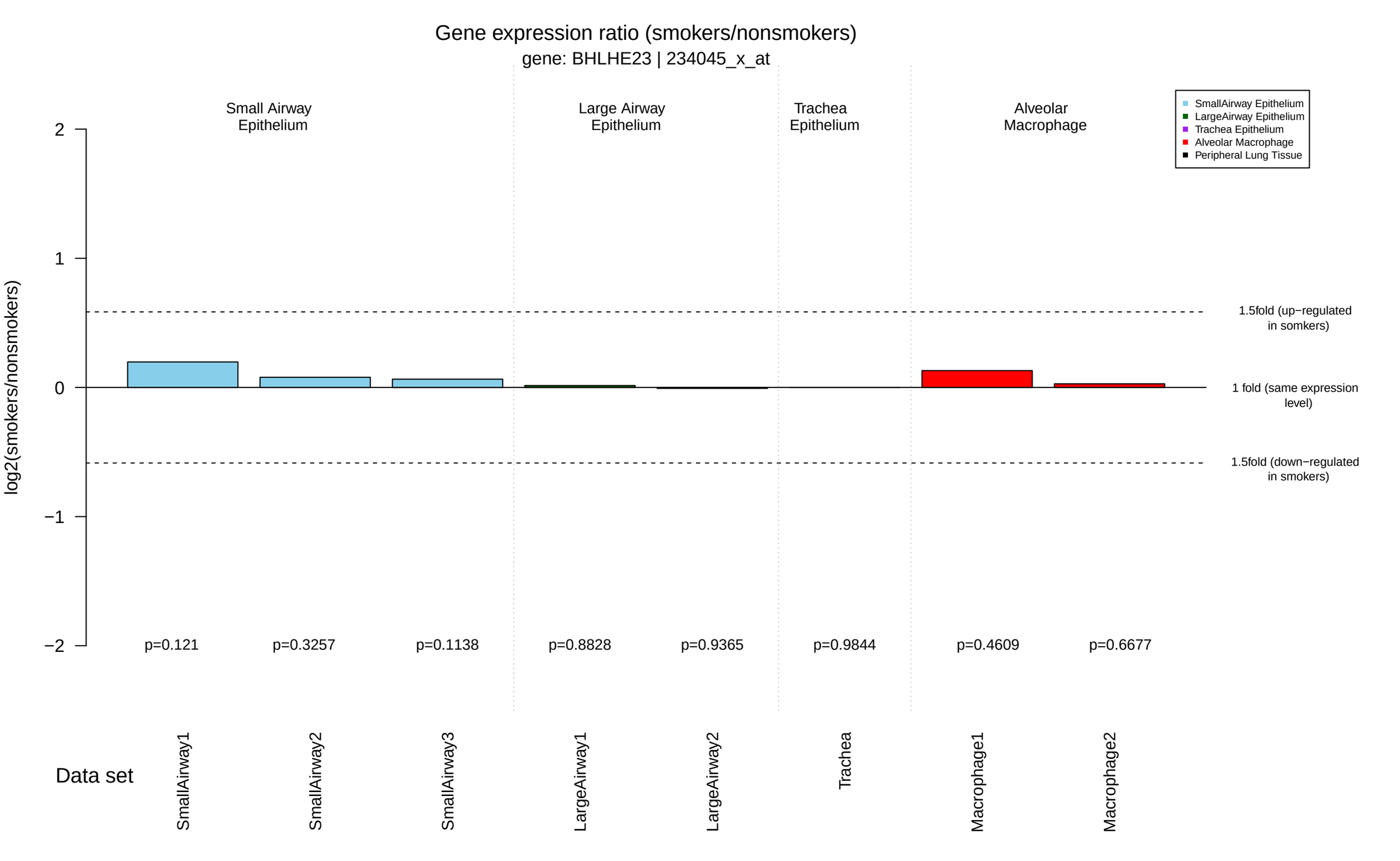
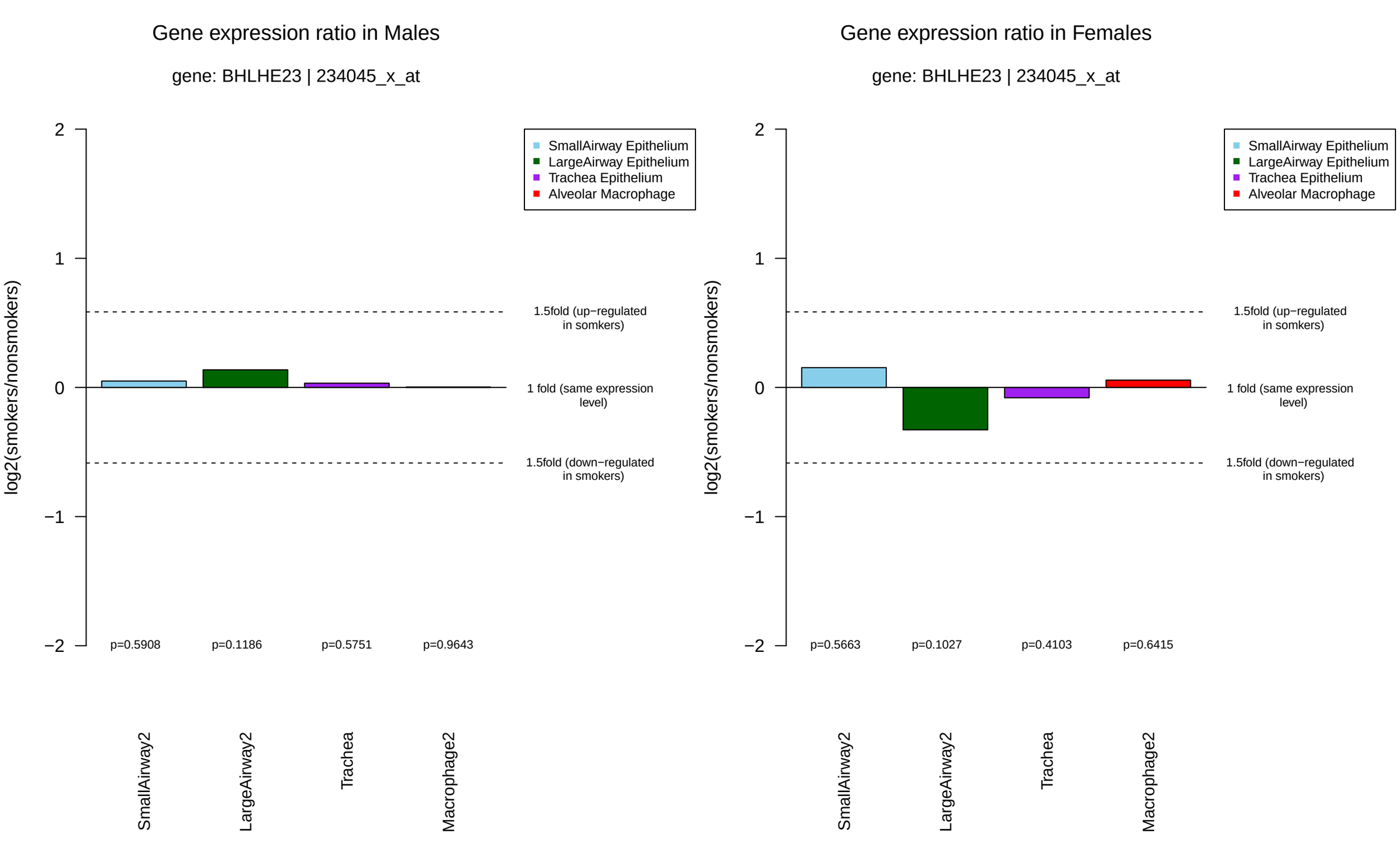
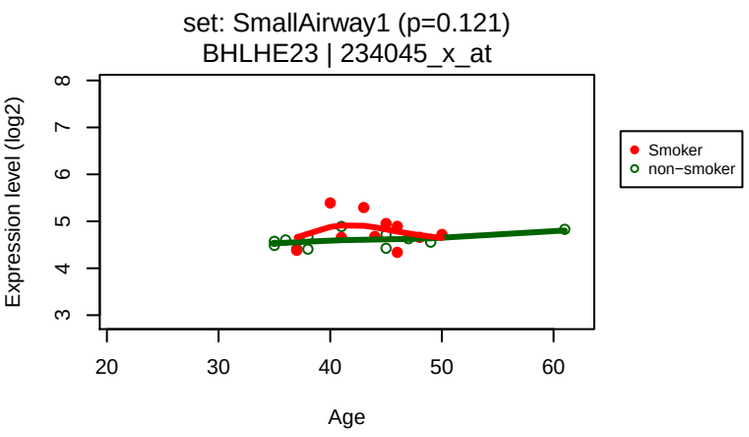
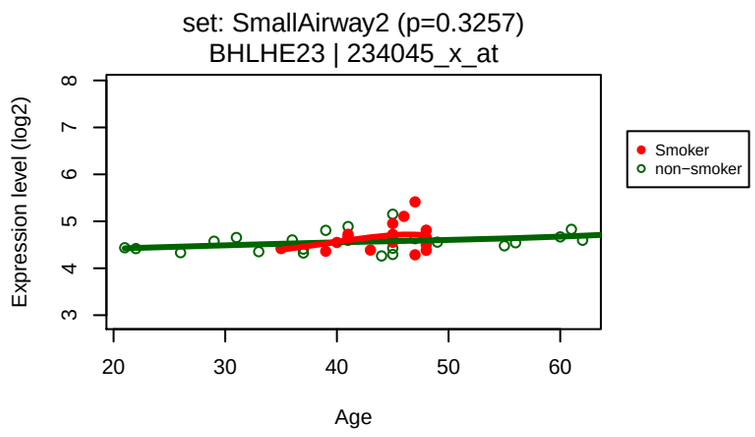
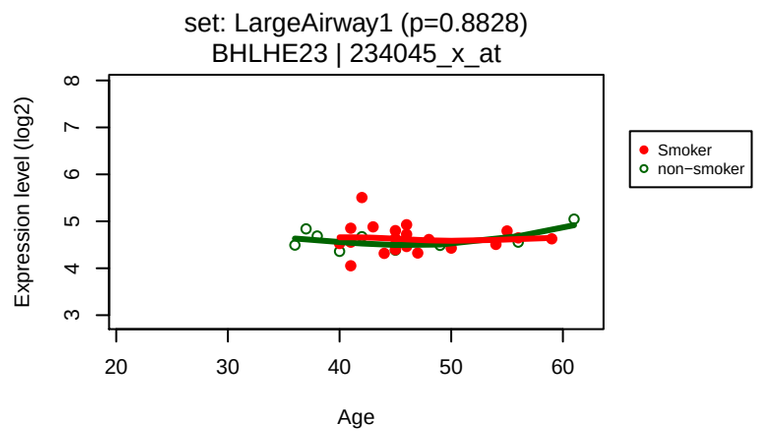

Gene expression of smokers and nonsmokers
gene: BHLHE23 | 234045_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.121
p=0.3257
p=0.1138
p=0.8828
p=0.9365
p=0.9844
p=0.4609
p=0.6677
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

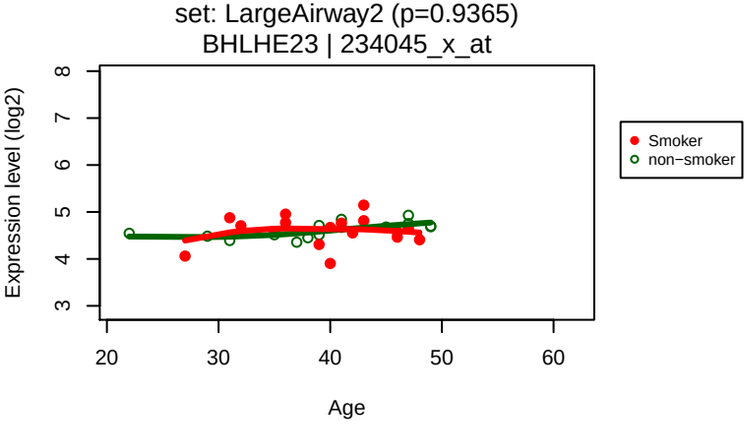

Gene expression ratio (smokers/nonsmokers)
gene: BHLHE23 | 234045_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.121
p=0.3257
p=0.1138
p=0.8828
p=0.9365
p=0.9844
p=0.4609
p=0.6677
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

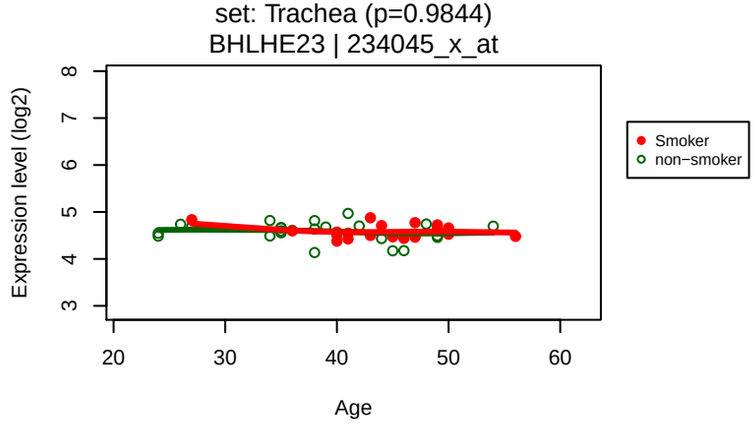

Gene expression ratio in Males
gene: BHLHE23 | 234045_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5908
p=0.1186
p=0.5751
p=0.9643
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BHLHE23 | 234045_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5663
p=0.1027
p=0.4103
p=0.6415
SmallAirway2
LargeAirway2
Trachea
Macrophage2

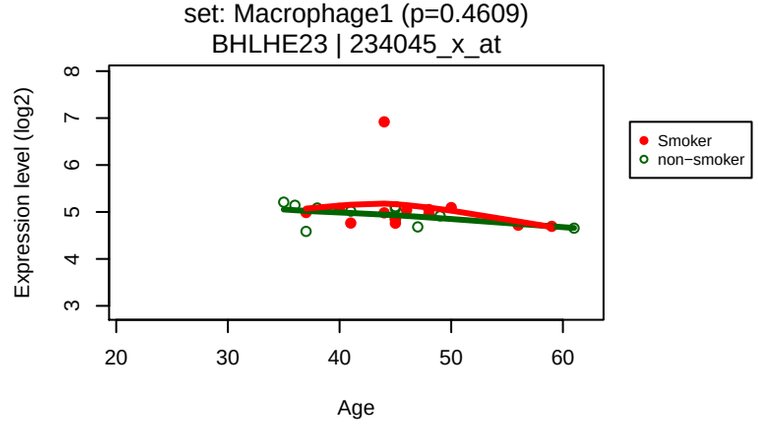

set: SmallAirway1 (p=0.121)
BHLHE23 | 234045_x_at
Expression level (log2)
Age
Smoker
non−smoker

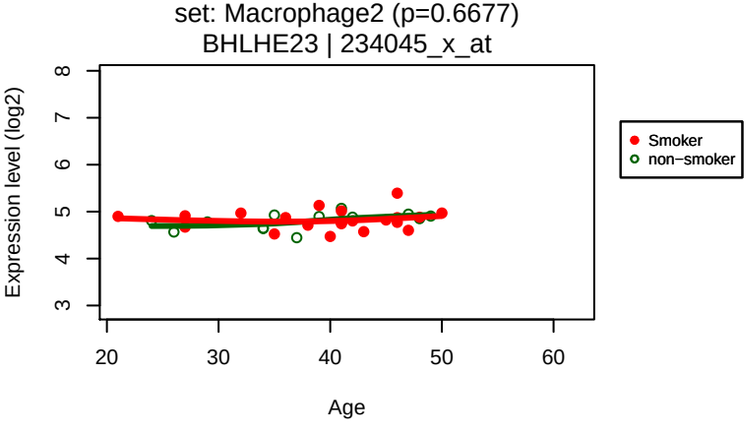

set: SmallAirway2 (p=0.3257)
BHLHE23 | 234045_x_at
Expression level (log2)
Age
Smoker
non−smoker

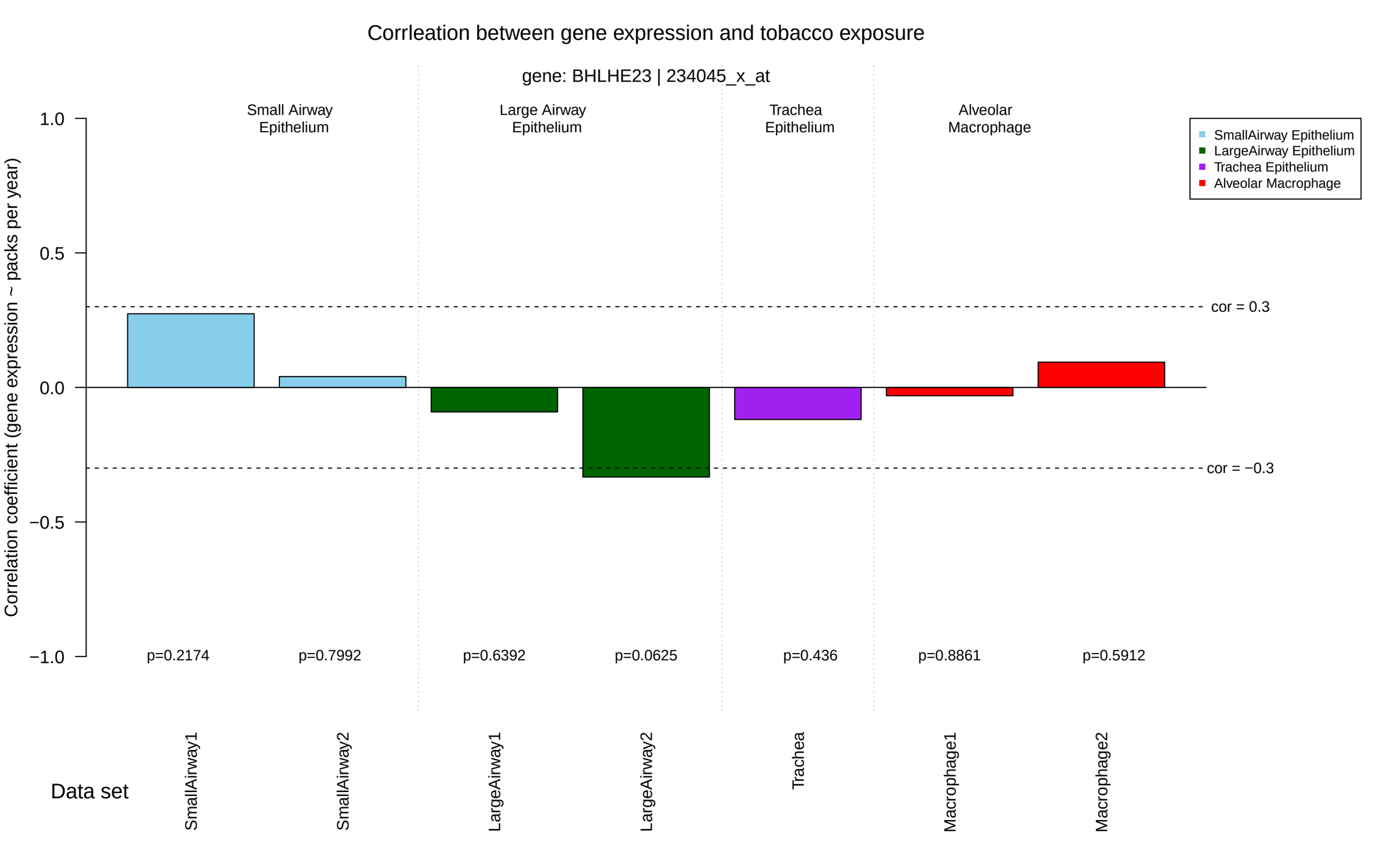

set: LargeAirway1 (p=0.8828)
BHLHE23 | 234045_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9365)
BHLHE23 | 234045_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9844)
BHLHE23 | 234045_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4609)
BHLHE23 | 234045_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6677)
BHLHE23 | 234045_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BHLHE23 | 234045_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2174
p=0.7992
p=0.6392
p=0.0625
p=0.436
p=0.8861
p=0.5912
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2